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Beyond majority: Label ranking ensembles based on voting rules

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ABSTRACT

Label ranking is a machine learning task that deals with mapping an instance to a ranking of labels, representing the labels' ordered relevance to the instance. Three recent studies have suggested the use of ensembles to improve the performance of simple label ranking models. However, none of them has explicitly inspected the question of how should the results obtained by simple models be aggregated into a single combined output. While classification tasks and regression tasks typically employ trivial aggregation techniques (i.e., majority voting and averaging, respectively), the case of label ranking tasks is not straightforward. The contribution of this paper is twofold. First, we propose to apply voting rules, typically used in the field of social choice, as the aggregation technique for label ranking ensembles. Our evaluation reveals that there is no single rule that consistently outperforms all other voting rules, and that under different settings different voting rules perform the best. Second, we propose a novel aggregation method for label ranking ensembles that learns the best voting rule to be used in a given setting. An extensive evaluation of the proposed method on semi-synthetic as well as real-world datasets shows that it obtains prediction performance that is significantly higher than that of the aggregation techniques currently used by state-of-the-art label ranking ensembles.

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1. Introduction

Label ranking has gained a lot of attention from the expert and intelligent systems community in recent years (Aiguzhinov, Soares, & Serra, 2010; Aledo, Gámez, & Molina, 2017; Cheng, Henzgen, & Hüllermeier, 2013; Cheng, Hühn, & Hüllermeier, 2009; Destercke, Masson, & Poss, 2015; Gurrieri, Fortemps, & Siebert, 2014; Hüllermeier, Fürnkranz, Cheng, & Brinker, 2008; Sá, Soares, Knobbe, & Cortez, 2017; Vembu & Gärtner, 2010; Zhou, Liu, Yang, He, & Liu, 2014; Zhou & Qiu, 2018). Label ranking is a prediction problem which deals with learning a mapping between an instance and a (possibly incomplete) ranking of labels from a finite set, representing their ordered relevance to the instance. Label ranking is the general case of *multi-class* and *multi-label* classification tasks (Har-Peled, Roth, & Zimak, 2003). In multiclass tasks an instance is associated with a single label out of k possible labels. This is equivalent to associating an instance with the top-one ranked label. Likewise, the top-k ranked labels provide a solution to the multi-label classification task.

While the importance of multi-class and multi-label problems is commonly understood, we provide a few motivating examples to the practical significance of label ranking tasks. First, as we show in the experimental section, label ranking can be used to predict the expression level of genes in a genome, where each gene has a phylogenetic profile (Balasubramaniyan, Hüllermeier, Weskamp, & Kämper, 2004). As another example, consider the problem of document categorization. A document collection such as news articles contains different topics such as sports, entertainment and politics. A document may belong to multiple topics, and the goal of the label ranking algorithm is to rank the relevant topics of the document (Dekel, Singer, & Manning, 2004). Lastly, in meta-learning, a label ranking model can provide a list of algorithms to a given problem, ranked according to their fit to the problem, based on the characteristics of the problem at hand (Brazdil, Soares, & Da Costa, 2003).

Ensemble models collect the results of multiple simple models and combine them into a single aggregated output. By building an ensemble, the mistake of one model can be compensated by the other models, and consequently, ensembles generally exhibit better performance than that of a single model (Dietterich, 2000; Kittler, Hatef, Duin, & Matas, 1998; Polikar, 2006; Rokach, 2016; Sagi & Rokach, 2018; Tax, Van Breukelen, Duin, & Kittler, 2000). Typically, ensembles can be broken into three sub-tasks: (1) choosing the base prediction algorithm, (2) generating simple models by

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repeatedly sampling the training data and training the base algorithm with the sampled data, and (3) aggregating the results obtained by the simple models into a single output.

Three recent studies (Aledo et al., 2017; Sá et al., 2017; Zhou & Qiu, 2018) have suggested the use of ensembles to improve the prediction performance of simple label ranking models. However, none of these studies has explicitly inspected the question of how should the results obtained by simple models be aggregated into a single output. Although majority voting makes perfect sense when aggregating categorical results (as commonly done in classification tasks) and averaging makes perfect sense when aggregating numerical results (as commonly done in regression tasks), they make far less sense in the case of rankings (for more details, see Section 3).

The contribution of this paper is twofold:

- **Voting rules matter:** To the best of our knowledge, this is the first study that explicitly investigates aggregation methods for label ranking ensembles. Thus far, only two aggregation techniques have been used (Aledo et al., 2017; Sá et al., 2017; Zhou & Qiu, 2018), while the field of social choice offers a wide variety of such aggregation techniques, known as voting rules. We therefore evaluate various voting rules for aggregating the results of simple label ranking models and compare their prediction performance over a large number of datasets (both semi-synthetic and real-world). We show that there exist voting rules that generally lead to better prediction performance than the two previously used rules. In addition, we demonstrate that there is no single rule that consistently outperforms all other voting rules, and that under different settings different voting rules perform the best.
- **Learning is powerful:** We propose a novel aggregation method for label ranking ensembles that learns the best voting rule to be used for a given dataset. An extensive evaluation of the proposed method shows that it obtains prediction performance that is significantly higher than that of the two aggregation techniques currently used by state-of-the-art label ranking ensembles. Furthermore, we show that being able to choose the right voting rule to be used for a given instance (rather than for the entire dataset) can lead to additional improvement in prediction performance.

The rest of this paper is organized as follows: we first discuss label ranking, ensembles, voting rules and label ranking ensembles in Section 2. Thereafter, we describe our proposed method in Section 3. This is followed by an overview of our experimental setting in Section 4, a description of our extensive evaluation and results in Section 5 and a summary and suggestions for future work in Section 6.

2. Related work

In this paper, we study aggregation techniques based on voting rules for label ranking ensembles. We therefore begin this section by providing the necessary background on three related fields of research: Label Ranking (Section 2.1), Ensembles (Section 2.2), and Voting Rules (Section 2.3). The section ends with a detailed description of the three existing state-of-the-art Label Ranking Ensembles (Section 2.4).

2.1. Label ranking

Label ranking is a prediction problem which deals with learning a mapping between an instance and an order (i.e., a ranking) of labels from a finite set, representing their relevance to the instance. As a motivating example, consider a college applicant that needs to decide what program she should apply to. A counselor can present a ranked list of suggestions, according to the applicant's profile, preferences, abilities and perhaps even the performance of past students. Thus, a suggestion to one applicant may be: math $\succ$ computer science $\succ$ biology $\succ$ medicine. As another example consider the problem of document categorization. A document collection such as news articles contains different topics such as sports, entertainment and politics. A document may belong to multiple topics, and the goal of the label ranking algorithm is to rank the relevant topics of the document.

Label ranking can be seen as the general case of *multi-class* and *multi-label* classification tasks. More specifically, in multi-class classification tasks, the goal is to predict the (single) correct class out of a number of possible classes. In multi-label tasks, the goal is to predict a few correct classes (the labels). In label ranking tasks, the goal is to predict the ranking of these labels. Note that the label ranking task should not be confused with the Learning to Rank (L2R) task (Cohen, Schapire, & Singer, 1998). In L2R problems, the goal is to produce a ranked set of instances (e.g., items) whereas in label ranking, the goal is to predict a ranked list of labels (from a finite and predetermined set of labels) for each instance.

Numerous label ranking algorithms were suggested in the literature. One approach is based on turning the problem into several binary classification problems and then combining them into output rankings. (Cheng et al., 2013; Destercke et al., 2015; Gurrieri et al., 2014; Har-Peled et al., 2003; Hüllermeier et al., 2008). Another common approach is based on modifying existing probabilistic algorithms to directly support label ranking. Some main examples are: naive Bayes models (Aiguzhinov et al., 2010), k-nearest neighbor models (Brazdil et al., 2003) and decision tree models (Cheng et al., 2009; de Sá, Rebelo, Soares, & Knobbe, 2015). For recent surveys on label ranking algorithms see Zhou et al. (2014) or Vembu and Gärtner (2010).

2.2. Ensembles

Ensemble models collect the results of multiple simple models and combine them into a single aggregated output. By building an ensemble, the mistake of one model can be compensated by the other models, and consequently, ensembles generally exhibit better performance than that of a single model (Dietterich, 2000; Kittler et al., 1998; Polikar, 2006; Rokach, 2016; Sagi & Rokach, 2018; Tax et al., 2000). Typically, ensembles can be broken into three subtasks:

- **Base algorithm:** choosing the base prediction algorithm for generating simple models. Depending on the prediction task at hand (i.e., classification, regression, label ranking, etc.), many base algorithms exist, and choosing the right algorithm may have a large impact on the overall performance of the ensemble.
- **Data sampling:** generating simple models by repeatedly sampling the training data and then training the base algorithm with the sampled data. The data must be sampled since if all models are trained with the same data, their output is likely to be identical and thus using an ensemble becomes meaningless. Several sampling approaches were suggested in the literature. The main ones include Bagging (where instances are sampled with replacement) (Breiman, 1996), Dagging (where instances are sampled without replacement) (Ting & Witten, 1997), Attribute Bagging (where attributes are sampled) (Barandiaran, 1998; Bryll, Gutierrez-Osuna, & Quek, 2003), Random Forest (where instances as well as attributes are sampled) (Breiman, 2001) and Boosting (where instances are sampled according to previous prediction errors) (Freund & Schapire, 1997). It is important to note that in the case of Boosting, data sampling and aggregation are interleaved.

- **Aggregation:** combining the results obtained by the simple models into a single aggregated output. Typically, aggregation is done with majority voting or averaging. While majority voting makes perfect sense when combining categorical results (as commonly done in classification tasks) and averaging makes perfect sense when combining numerical results (as commonly done in regression tasks), they make far less sense in the case of rankings. In the next section we describe voting rules that can be used for aggregating the results of label ranking models. Another option for aggregation is to employ Stacking (Gama & Brazdil, 2000; Wolpert, 1992). That is, building a meta-classifier which uses the outputs of the various simple models as input attributes, in order to predict a single aggregated output.

Ensemble methods were extensively studied in the cases of classification tasks (Breiman, 2001), multi-label tasks (Tsoumakas & Vlahavas, 2007), regression tasks (Bühlmann & Yu, 2003), learning to rank tasks (Burges, Svore, Bennett, Pastusiak, & Wu, 2011; Freund, Iyer, Schapire, & Singer, 2003) and even unsupervised learning tasks (Bacher, Ben-Gal, & Shmueli, 2017; Parisi, Strino, Nadler, & Kluger, 2014). However, ensembles were hardly studied in the context of label ranking tasks as we further describe in Section 2.4.

2.3. Voting rules

In the domain of social choice, voting rules are used to combine individual opinions, preferences, or interests to reach a collective decision (Brams & Fishburn, 2002). Voting rules are commonly used in democratic election settings, ranging from national elections to head of school parent council elections.

More formally, let there be n voters, $V = \{v_1, v_2, \ldots, v_n\}$ and m candidates $C = \{c_1, c_2, \ldots, c_m\}$. Each voter v_i ranks the candidates according to his or her preferences, creating a preference profile $\pi \in \Omega$ where Ω is the set of all possible preference profiles (the permutation space). We write $c_i \succ c_j$ when c_i is preferred over c_j. For example, the preference profile π_i of voter v_i might be: $c_2 \succ c_1 \succ c_3$. Let $\Pi = \{\pi_1, \ldots, \pi_n\}$ denote the set of preference profiles of all voters and $N(c_i, c_j)$ denote the number of profiles in Π in which $c_i \succ c_j$. A voting rule r receives the set of preference profiles Π and outputs a single ranked list of items π_r.

Voting rules differ in the manner they compute the output π_r. Unfortunately, Arrow's impossibility theorem states that there is no voting rule that can satisfy a few simple axioms (Arrow, 2012). This result in some way resembles the no-free-lunch theorem that proves that a classifier that is the best for all classification problems does not exist (Wolpert & Macready, 1997). Next, we provide a non-exhaustive list of commonly used voting rules:

- Plurality Ranking. The score of each candidate c_i is computed as the number of profiles in Π in which c_i was ranked first. Candidates are then ranked according to their scores.
- Veto. The score of each candidate c_i is computed as the number of profiles in Π in which c_i was not ranked last. Candidates are then ranked according to their scores.
- Single Transferable Vote (STV). π_r is computed in an iterative process: the first place goes to the candidate ranked first the most times. This candidate is then removed, and the second place goes to the candidate who is now ranked first the most times. The process continues in this manner until all candidates are ranked.
- Maximin. The score of candidate c_i is the minimum sum of victories in pairwise comparisons: $min(_{\forall j \neq i} N(c_i, c_j))$. Candidates are ranked according to their scores (Young, 1977).
- Modal Ranking. The mode (most frequent) ranking in Π is selected as π_r (Caragiannis, Procaccia, & Shah, 2014).

Table 1
Label ranking ensemble frameworks.

	Aledo et al. (2017)	Sá et al. (2017)	Zhou and Qiu (2018)
Base algorithm	LRT	ERT, RT	TLAC
Data sampling	Bagging	Random Forest	Random Forest
Aggregation	Modal ranking	Borda	Borda

- Borda. The score of candidate c_i is $\sum_{j=1, j\neq i}^{m} N(c_i, c_j)$. Candidates are then ranked according to their scores (de Borda, 1781).
- Copeland. The score of candidate c_i is the sum of victories in pairwise comparisons: $\sum_{j=1, \forall j\neq i}^{m} [N(c_i, c_j)) > n/2]$. Candidates are ranked according to their scores (Copeland, 1951).

Borda was shown to work well in different settings (Ali & Meilă, 2012), and indeed is widely used in ranking problems of various kinds. In information retrieval, as part of the learning to rank problem (Liu, 2011), Borda is used for meta-search engines (Aslam & Montague, 2001; Carterette & Petkova, 2006). In recommender sytems, Borda was used to aggregate preferences from multiple users (Naamani-Dery, Golan, Kalech, & Rokach, 2015). Borda and Copeland have also been used to create a consensus cluster from several clusters, and have exhibited better performance than simple methods such as product or sum (Sevillano, Alías, & Socoró, 2012). Copeland may outperform Borda in ranking web results (Dwork, Kumar, Naor, & Sivakumar, 2001). Lastly, Van Erp, Vuurpijl, and Schomaker (2002) experimented with a few voting methods including Borda and Plurality on two datasets of pattern recognition, using Bagging. However their objective was to compute one class (i.e. pattern) and not to output a ranked order over the classes therefore, apart from Borda, all the other voting rules they examine cannot (and need not) output a ranking.

In the above studies voting rules were often used to aggregate preferences in settings where there is no clear ground-truth decision, i.e., the correct decision is interpreted differently by voters with different views of what is right. Nevertheless, votes can be seen as a noisy estimate of some underlying truthful ranking (De Condorcet, 1785; Young, 1995). Finding the best voting rule to estimate the truth in different noisy settings has gained a lot of interest recently (Caragiannis, Procaccia, & Shah, 2016; Mao, Procaccia, & Chen, 2013; de Weerdt, Gerding, & Stein, 2016). Mao et al. (2013) used synthetic data from the Mallows noise model, as well as data from a case study limited to ranking sets of four items. They observed which rules output a ranking that has the smallest distance from the true ranking. They found that Borda is a good and simple rule to use. In a similar study, de Weerdt et al. (2016) suggested Copeland should be used.

2.4. Label ranking ensembles

To the best of our knowledge, only three papers thus far have investigated the use of ensembles for label ranking (Aledo et al., 2017; Sá et al., 2017; Zhou & Qiu, 2018). The ensembles proposed in these papers differ in: (1) the base label ranking algorithm used, (2) the method used to sample the data to train each of the simple models (if all models are trained with the exact same data they will output the exact same results, and then there is no need for an ensemble), and (3) the aggregation method used to combine the results of the simple models. These three papers and the main differences between them are discussed next and are summarized in Table 1.

As the base label ranking algorithm Aledo et al. (2017) used Label Ranking Trees (LRT) (Cheng et al., 2009; Sá et al., 2017) used Ranking Trees (RT) and Entropy Ranking Trees (ERT) (de Sá et al., 2015) and (Zhou & Qiu, 2018) developed their own method named Top Label As Class (TLAC).

Table 2
Example of voters' preferences.

v_1	v_2	v_3	v_4	v_5	v_6	v_1	v_2	v_3	v_4	v_5	v_6
c_1	c_1	c_1	c_2	c_2	c_3	c_1	c_1	c_1	c_2	c_2	c_3
						c_2	c_2	c_2	c_3	c_3	c_2
						c_3	c_3	c_3	c_1	c_1	c_1

To select the training data for each simple classifier, (Aledo et al., 2017) used a technique known as Bootstrap aggregation or Bagging (Breiman, 1996): they created b different bags ($b = 5, 10.100$) by selecting a subset of the dataset's instances with replacement. The other two papers (Sá et al., 2017; Zhou & Qiu, 2018) suggested modifications to the well-known Random Forest ensemble model (Breiman, 2001). In Random Forest ensembles, as in Bagging, a subset of the dataset's instances is sampled with replacement. However, differently from Bagging, in Random Forest, a subset of the m features is also sampled ((Sá et al., 2017) selected $\sqrt{m}$ features whereas (Zhou & Qiu, 2018) selected $\log_2 m$ features).

As for the aggregation method used, (Aledo et al., 2017) used what they termed as "majority voting" in order to aggregate the results. After attempting to replicate their results, we believe that they actually used Modal Ranking (see Caragiannis et al. (2014) and Section 2.3). Sá et al. (2017) stated that they computed the average rank position of each label and ordered the labels according to their average. This is equal to combining the rankings using the Borda voting rule (de Borda, 1781), which computes the sum of the rank positions for each label and orders the labels according to their sum. Zhou and Qiu (2018) explicitly stated that they use the Borda voting rule to combine the rankings of the classifiers.

3. The proposed method: Label ranking ensembles based on voting rules

The problem of aggregating the results of several simple label ranking models into a single result is not trivial, as we demonstrate in the following example. Let there be six voters $v_1, v_2, \ldots, v_6$ and three candidates c_1, c_2, c_3 as shown in Table 2.

The left part of Table 2 shows the case where each voter reported her top-1 candidate, as done in multi-class classification tasks. In this case, assuming that all voters are treated equally, choosing the most frequent preference (c_1), formally termed plurality voting, makes a lot of sense.

Now, consider the right side of Table 2, where each voter provides her preferences for all candidates, e.g., voters v_1, v_2 and v_3 share the same preference which is: $c_1 \succ c_2 \succ c_3$. In this case, although c_1 is the most preferred candidate for three of the six voters, it is the least preferred candidate for the other three voters. In this case, it makes much less sense that c_1 will be chosen as the winner. Moreover, in many cases, we are interested in outputting a complete ranking of the candidates and not only a single winner.

As described in Section 2.4, previous studies on label ranking ensembles have (implicitly) used Modal Ranking and Borda for aggregating the results of the simple label ranking models. In the example above, using Modal Ranking will lead to the final aggregated ranking of $c_1 \succ c_2 \succ c_3$ and using Borda will lead to the final aggregated ranking of $c_2 \succ c_1 \succ c_3$.

We hypothesize that: (1) there exist other voting rules that can lead to better prediction performance than the two already used voting rules, (2) different voting rules will excel in different settings, and (3) it is possible to learn (and therefore predict) which voting rule will perform best in which setting.

Based on these hypotheses, we propose a novel aggregation method for label ranking ensembles, Voting Rule Selector (VRS), that learns the best voting rule to be used in a given setting. A pseudo-code of the algorithm is provided in Algorithm 1 and a detailed description follows.

Algorithm 1 Voting Rule Selector (VRS).

1: **Input:**
   A set of simple label ranking models: $m_1, m_2 \ldots, m_M$
   A set of validation instances V, with rankings as targets
   A set of voting rules A
2: $\bar{s} \leftarrow []$
3: **for** $a \in A$ **do**
4: $\quad s \leftarrow []$
5: $\quad$ **for** $v \in V$ **do**
6: $\quad\quad \pi \leftarrow a(m_1(v), \ldots, m_M(v))$
7: $\quad\quad s[v] \leftarrow kt(\pi, target(v))$
8: $\quad$ **end for**
9: $\quad \bar{s}[a] \leftarrow \frac{1}{|V|} \sum_{v \in V} s[v]$
10: **end for**
11: $\hat{a} \leftarrow \underset{a \in A}{\operatorname{argmax}}\ \bar{s}[a]$
12: **Output:**
   $\hat{a}$ - the best performing aggregation technique over V

The algorithm receives as input a set of M simple label ranking models, a set of validation instance V, and a set of voting rules A (line 1). The algorithm iterates over each voting rule a in A (line 3) and over each instance v in V (line 5). In each iteration, the algorithm applies each of the simple models over v to obtain a single ranking, and aggregates these ranking using the voting rule a (line 6). Then, the algorithm compares the aggregated ranking and the actual target ranking (as appears in V) using the Kendall-tau (KT) coefficient function (Kendall, 1938; 1962). KT can be seen as a similarity function between two rankings which is based on counting the minimal number of swaps needed to transform one ranking into the other[1] (line 7). Then, the average KT value over all validation instances is calculated (line 9). Finally, the voting rule that performed best (i.e., the one that obtained the highest average KT value) is returned (lines 11–12).

While VRS can potentially be used as the aggregation technique when using various data sampling techniques and base learning algorithms (see Section 2.2 for more details), we focus on its usage with Bagging (and an arbitrary base learning algorithm). This usage is formalized in Algorithm 2 and is described next.

The algorithm receives as input a set of train instances D_{train}, a set of test instances D_{test}, a base label ranking algorithm *weak_learner*, the number of bags B, the train-validation ratio R, and a set of voting rules A (line 1). The algorithm first splits the train set D_{train} into a train set T and a validation set V, according to the given ratio R (line 2). The algorithm proceeds by sampling B bags and training the base label ranking algorithm *weak_learner* with each bag, resulting in B simple models (lines 3–6). Then, VRS is executed to select the voting rule to be used as the aggregation technique (line 7). Finally, the average accuracy of using the chosen voting rule over the test set D_{test} is calculated (lines 8–13) and returned (line 14).

4. Experimental setting

In this section, we describe the experimental flow, the datasets used and the compared aggregation methods.

[1] We use the Kendall-tau-b variation which can handle ties in the order of rankings.

Algorithm 2 Bagging with VRS (BVRS).

Input:
A set of train instances D_{train} with rankings as targets
A set of test instances D_{test} with rankings as targets
A base label ranking algorithm *weak_learner*
The number of bags B
The train-validation ratio R
A set of voting rules A
$T, V \leftarrow split(D_{train}, R)$
for $b \in 1, \ldots, B$ **do**
  $S \leftarrow$ sample $|T|$ instances from T with replacement
  $m_b \leftarrow weak_learner(S)$
end for
$a \leftarrow VRS(m_1, \ldots, m_B, V, A)$
$s \leftarrow []$
for $i \in D_{test}$ **do**
  $\pi \leftarrow a(m_1(i), \ldots, m_B(i))$
  $s[i] \leftarrow kt(\pi, target(i))$
end for
$\bar{s} \leftarrow \frac{1}{|D_{test}|} \sum_{i \in D_{test}} s[i]$
Output:
$\bar{s}$ - The average accuracy over instances in D_{test}

4.1. Experimental flow

Our experimental flow follows a similar approach to that of Aledo et al. (2017) and is composed of four main steps which are illustrated in Fig. 1:

1. **Cross validation.** Given a label ranking dataset (see list of datasets below), we employed a 10-fold cross-validation process, in which the dataset was partitioned into ten distinct folds, and then in ten iterations, one fold was used as a test set and the remaining were used for the training set (Step A).
2. **Bagging and training.** Given a training set, multiple bags were sampled, and each such bag was used to train a single label ranking prediction model (Step B). For our evaluation, we used 50 bags and the LRT label ranking base algorithm. Aledo et al. showed in their paper that the *LRT* algorithm with Bagging outperforms other state-of-the-art label ranking methods. Moreover, they showed that there is no significant improvement when using more than 50 bags. Our own experiments yielded consistent results. To create sufficient diversity between the bags, each bag was the result of sampling (with replacement) 30% of the training instances.
3. **Prediction.** Each one of the test instances was entered as input to each one of the 50 trained models, and the 50 outputted rankings were aggregated into a single aggregated output, using one of the aggregation techniques described below (Step C).
4. **Evaluation.** The aggregated ranking was compared to the real ground-truth ranking using the Kendal-tau-b coefficient (Step D). The resulting coefficients were averaged over all test instances, and then over the ten test folds, to produce a single measure for a dataset.

Steps A and B of the evaluation were implemented in Java 9.0.1, whereas the rest of the steps were implemented in Python 2.7.

4.2. Datasets

We used as benchmark a total of 24 datasets. Table 3 shows the main characteristics of each dataset.

The first 16 datasets were proposed by Cheng et al. (2009) and are used since then as a standard benchmark for the label ranking problem. They can be considered semi-synthetic, as they were obtained by transforming a multi-class or regression problem to a label ranking one (see Cheng et al. (2009) for the transformation details).

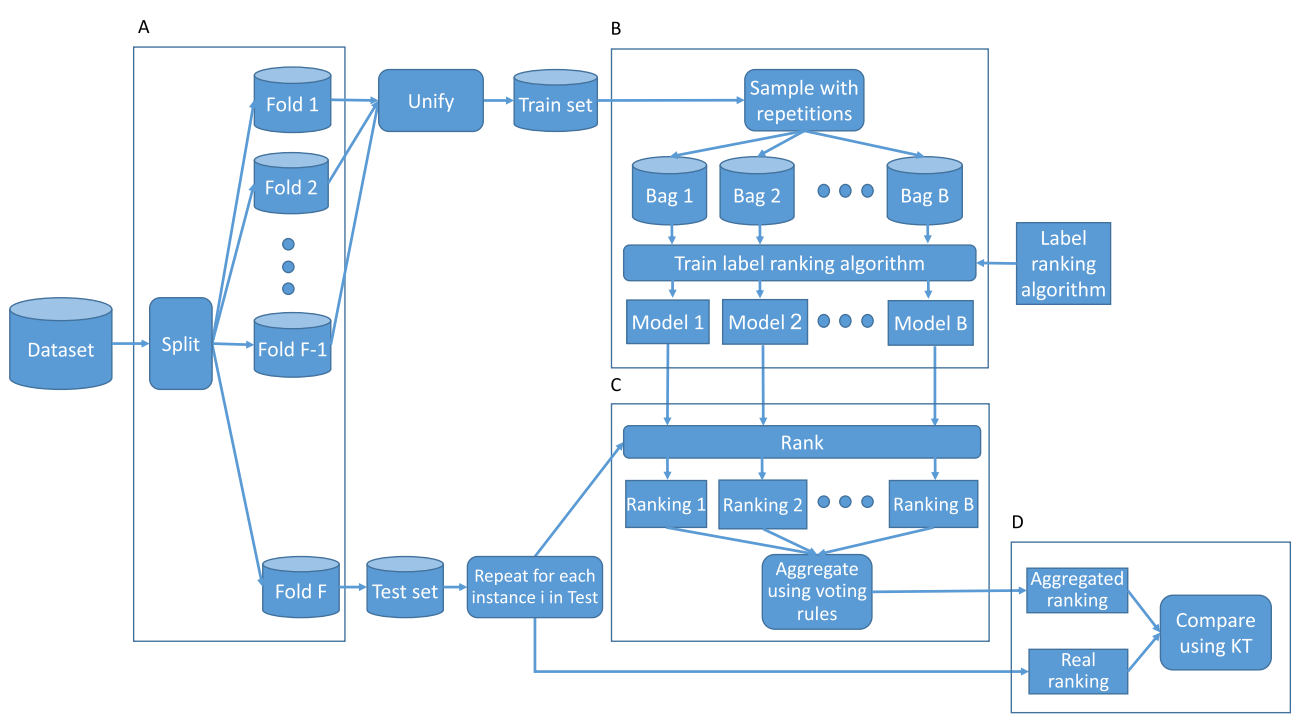

Fig. 1. The experimental flow.

Table 3
Datasets characteristics.

Dataset name	Type	# of instances	# of features	# of labels
Authorship	Semi-synthetic	841	70	4
Bodyfat	Semi-synthetic	252	7	7
Calhousing	Semi-synthetic	20,640	4	4
Cpu-small	Semi-synthetic	8192	6	5
Elevators	Semi-synthetic	16,599	9	9
Fried	Semi-synthetic	40,760	9	5
Glass	Semi-synthetic	214	9	6
Housing	Semi-synthetic	506	6	6
Iris	Semi-synthetic	150	4	3
Pendigits	Semi-synthetic	10,992	16	10
Segment	Semi-synthetic	2310	18	7
Stock	Semi-synthetic	950	5	5
Vehicle	Semi-synthetic	846	18	4
Vowel	Semi-synthetic	528	10	11
Wine	Semi-synthetic	178	13	3
Winsconsin	Semi-synthetic	194	16	16
Cold	Real-world	2465	24	4
Diau	Real-world	2465	24	7
Dtt	Real-world	2465	24	4
Heat	Real-world	2465	24	6
Spo	Real-world	2465	24	11
German2005	Real-world	402	31	5
German2009	Real-world	407	33	5
Sushi	Real-world	5000	10	10

The last eight are real-world datasets. Five of them correspond to real-world biological data problems and were used by Hüllermeier et al. (2008). The labels in these datasets represent the expression level of genes of the yeast genome, where each gene has a phylogenetic profile of length 24 (i.e., 24 attributes). In this case the expression profile of the output gene (e.g. (1.7, 2.9, 0.3, −2.4)) was directly converted into a rank (e.g. (2, 1, 3, 4)). The other three datasets were used by de Sá et al. (2018). One dataset contains demographics and preferences over sushi types (Kamishima, 2003) and the other two contain social-economic and electoral results from Germany (Boley, Mampaey, Kang, Tokmakov, & Wrobel, 2013).

Since this study considers the case of complete rankings only, we had to remove 10 instances from the german2005 dataset and 5 instances from the german2009 dataset, which contained partial rankings in their target variable. The other 22 datasets contained complete rankings in all of their instances and were therefore remained untouched.

All 24 datasets, conveniently formatted for the label-ranking problem, can be downloaded from https://cs.uni-paderborn.de/?id=63912 and https://doi.org/10.17632/3mv94c8jpc.2.

4.3. Aggregation methods

We compared the prediction performance of a simple **single model** for label ranking based on *LRT*, and 10 different label ranking ensembles as described next.

Seven of the label ranking ensembles differed only by the voting rule used for aggregation. Specifically, we used the following voting rules: **Plurality, Veto, STV, MaxiMin, Modal Ranking, Borda** and **Copeland**. Recall that Modal Ranking and Borda are the rules currently used in state-of-the-art ensembles for label ranking (see Section 2.4).

In addition to the seven ensemble methods described above, we also implemented and evaluated two methods that can serve as upper bounds to learning abilities: **Oracle (Dataset)** - a method that chooses the best performing voting rule for each dataset, and **Oracle (Instance)** - a method that chooses the best performing voting rule for each instance. It is important to note that these two methods utilize information on the target variable and therefore cannot be used for prediction in practice. We therefore name these methods "oracles" and use them as upper bounds for learning only.

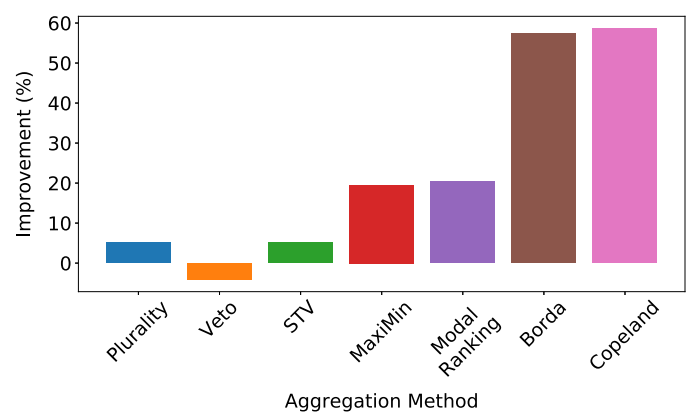

Fig. 2. Average improvement for ensembles based on the seven voting rules in comparison to the single model approach.

Finally, we implemented the **VRS** algorithm described in Section 3 that learns the best voting rule to be used for each dataset.

5. Results

In the first experiment, we wanted to compare the prediction performance of the seven voting rules (out of which Modal Ranking and Borda are the rules used in state-of-the-art ensembles for label ranking). Fig. 2 reports the average improvement (in percentages) of each of the voting rules in comparison to the single model approach. To do so, we first calculated the average Kendall-Tau-b for each rule (first over all instances in a given dataset, and then over all 24 datasets) and then calculated the improvement in percentages in comparison to the simple model approach. Inspecting Fig. 2, we observe the following:

- **The power of ensembles:** with the exception of Veto, all of the ensemble methods performed better than the single model

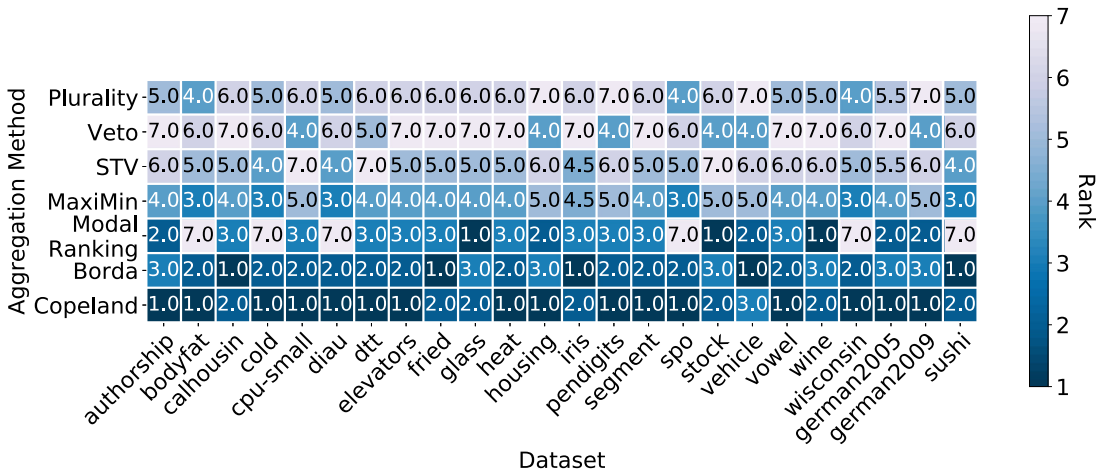

Fig. 3. Rank of voting rules ensembles for each of the 24 dataset.

approach. For example, Copeland obtained a Kendall-tau-b score that was higher than that of the single model in 58.6%. This observation emphasizes the superiority of ensemble methods over simple models in general, and in the case of label ranking tasks in particular.

- **Beyond majority:** among all seven voting rules, the three variations of Plurality: Plurality, Veto and STV, obtained the poorest improvement results (5.1%, −4% and 5.2%, respectively). In fact, Veto performed even worse than the single model approach. This demonstrates again why label ranking ensembles should be treated differently than classification ensembles.
- **The first place goes to Copeland:** existing state-of-the-art label ranking ensembles use either Modal Ranking (Aledo et al., 2017) or Borda (Sá et al., 2017; Zhou & Qiu, 2018). Interestingly, the voting rule that performs best is actually Copeland with an average improvement of 58.6% (compared to 20.5% and 57.4% obtained by Modal Ranking and Borda, respectively).

While the figure above showed the average improvement over all datasets, Fig. 3 provides a breakdown of the 24 datasets, where each entry represents the rank of the given voting rule for the given dataset (a rank of 1 means that the given voting rule performed the best for the given dataset, and a rank of 7 means that it performed the worst).

As can be observed from Fig. 3, **there is no single best performing rule** across all datasets. While Copeland was ranked first in 16 of the datasets, Borda was the best performer in five of the datasets and Modal Ranking in three of the datasets. This confirms our hypothesis that different rules may perform better in different settings.

In the second experiment, we were interested in testing whether it is possible to learn the best voting rule to be used for each dataset. Fig. 4 (top) shows the average improvement in percentages for each one of the seven ensembles based on voting rules as well as for the VRS algorithm and the Oracle (Dataset) method. Fig 4 (center) shows the percentage of datasets for which each aggregation method performed the best. Fig 4 (bottom) shows the average rank for each of the aggregation methods. As can be seen from the figure:

- **Learning at the dataset level is valuable:** the VRS method obtained an average improvement of 58.8%, which is slightly better than the best single voting rule - Copeland with 58.6%. It also achieved the best improvement in 79.1% of the datasets (in comparison to 66.7% in the case of Copeland), and obtained an average rank of 1.5 (in comparison to an average rank of 2 in the case of Copeland). This demonstrates that there is an advantage to learning the best rule to be used with each dataset, instead of selecting an arbitrary rule.
- **Enhancing state-of-the-art methods:** the VRS method is also remarkably better than the two rules used by existing state-of-the-art label ranking ensembles (Modal Ranking in Aledo et al. (2017) and Borda in Sá et al. (2017); Zhou and Qiu (2018)).
- **Glass ceiling at the dataset level:** According to the dataset-level oracle, carefully choosing the voting rule to be used for each dataset results in an improvement of 58.8% in comparison to the single model approach. This result is only slightly higher than that obtained by Copeland, meaning there is a very slim margin for potential improvement.

In the third experiment, we were interested in testing whether it is possible to obtain even better prediction performance by choosing the right voting rule to be used for each instance. Fig. 5 (top) shows the average improvement in percentages for each one of the seven ensembles based on voting rules as well as for VRS, Oracle (Dataset) and Oracle (Instance) methods. Fig. 5 (bottom) shows the percentage of instances for which each aggregation method performed the best.

By looking at Fig. 5, we observe that **choosing the right voting rule at the instance level can lead to additional improvement.** More specifically, the results obtained for Oracle (Instance) show that carefully choosing the voting rule to be used for each instance can lead to further improvement in prediction accuracy - 187.3% improvement in comparison to the single model approach.

Finally, we were interested in testing how the number of simple models used in the ensemble affect prediction performance. Fig. 6 shows the improvement in percentages as a function of the number of bags used for each of the aggregation methods.

As can be seen in Fig. 6, **the prediction performance grows (with a diminishing effect) with the number of simple models used.** More specifically, it seems that the VRS method outperforms the rest of the methods for all considered number of bags values. Moreover, when using 50 bags, prediction performance seems to stabilize for all of the considered aggregation methods.

In order to complete the picture, we performed two sets of statistical significance tests to compare between the 10 examined aggregation methods. Each of the two sets of tests followed the robust non-parametric procedure described in Demšar (2006); García, Fernández, Luengo, and Herrera (2010). First, we used the Friedman Aligned Ranks test in order to reject the null hypothesis

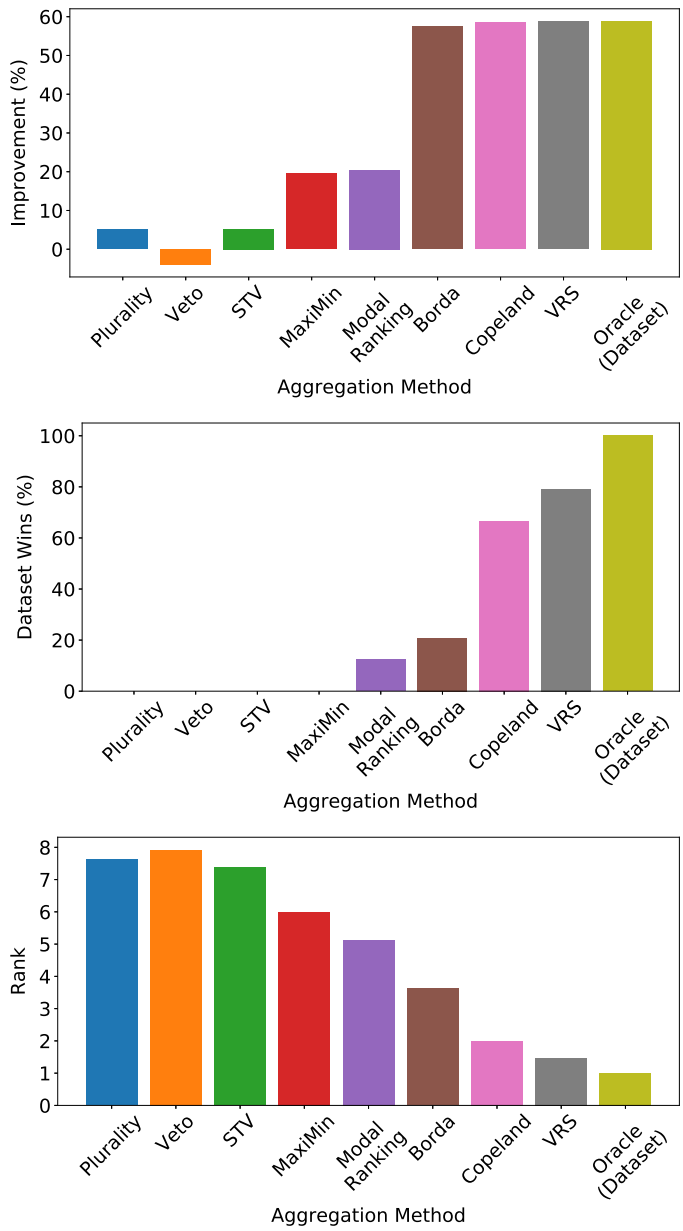

Fig. 4. Learning at the dataset level.

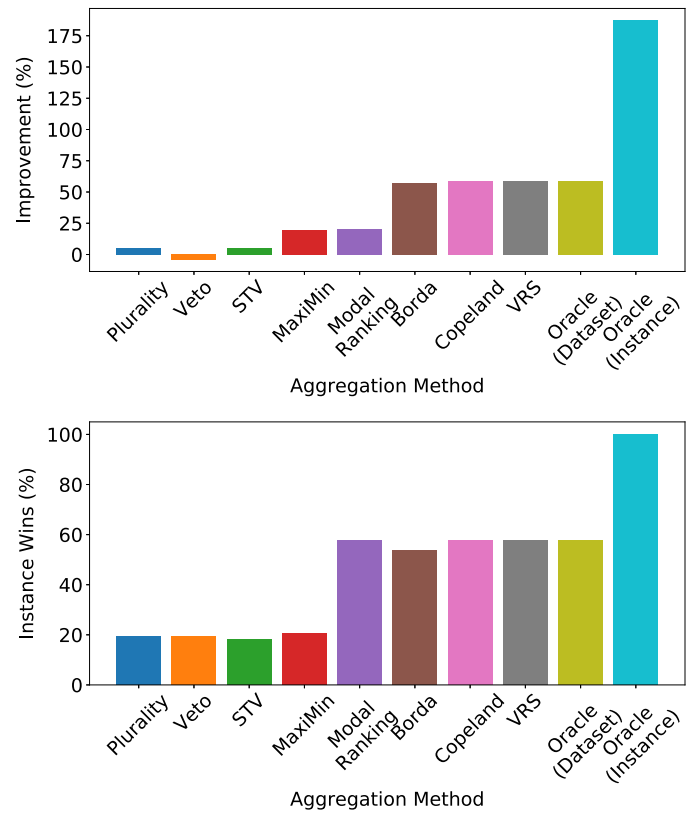

Fig. 5. The potential of learning at the instance level.

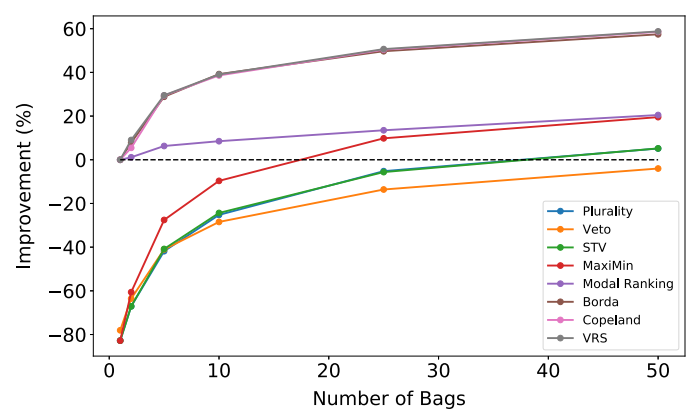

Fig. 6. Improvement in percentages as a function of the number of bags used.

that all methods in the set perform the same. Then, we used the Conover post hoc test to perform pairwise comparison between the various methods in the set.

In the first set of tests, we compared the seven aggregation methods based on single voting rules. The Friedman Aligned Ranks test with a confidence level of 95% rejected the null-hypothesis that all seven methods perform the same. Further applying the Conover post hoc test supported our finding above: Copeland performed significantly better than the other six method, including Modal Ranking and Borda.

In the second set of tests, we compared the following four aggregation methods of interest: Copeland, VRS, Oracle (Dataset) and Oracle (Instance). The Friedman Aligned Ranks test with a confidence level of 95% rejected the null-hypothesis that all four methods perform the same. Further applying the Conover post hoc test supported our findings above: VRS performed significantly better than Copeland; Oracle (Dataset) performed significantly better than VRS; and Oracle (Instance) performed significantly better than Oracle (Dataset).

All of the analyses reported in this section were repeated using the Ranking by Pairwise Comparison (RPC) algorithm, suggested by Hüllermeier et al. (2008), as the base label ranking learner, instead of the LRT algorithm which was used throughout this section. Our findings in the case of RPC were completely consistent with those of LRT and are described in detail in the appendix.

6. Conclusions and future work

In this paper we studied the aggregation sub-task of label ranking ensembles. We first suggested to use voting rules as the aggregation technique, and showed that there exist voting rules that generally lead to better prediction performance than the two previously used rules. Moreover, we demonstrated that under different settings different voting rules perform the best. Then, we pro-

posed a novel aggregation method, VRS, that learns the best rule to be used for a given dataset. An extensive evaluation of the proposed method showed that it significantly outperforms all examined voting rules, including the two rules previously used by state-of-the-art label ranking ensembles.

The VRS algorithm proposed in this paper can be quite easily incorporated in every setting which involves label ranking ensembles to improve their overall prediction performance. However, the studied approach of learning the best aggregation technique to be used in a given setting is not limited to label ranking ensembles, and may also be valuable in a variety of other fields that are concerned with aggregation of rankings, including: rank aggregation in information retrieval (Dwork et al., 2001), group recommendations in recommender systems (Ricci, Rokach, & Shapira, 2015), combining voters' preferences in computational social choice (Chevaleyre, Endriss, Lang, & Maudet, 2007), phylogenetic profiling (Balasubramaniyan et al., 2004), and even error correction codes (Procaccia, Shah, & Zick, 2016).

It is important to note two inherent limitations of the proposed VRS algorithm. First, VRS requires a fraction of the training data for validation, which means less data for training the base models is available. However, as we saw throughout our evaluation, this did not prevent VRS from outperforming the other methods even in the case of small datasets. Second, VRS requires execution of all voting rules during its training phase, which may result in increased runtime on very large datasets. However, this is not a real issue as the training phase is executed only once (or once in a period).

Future research directions may include the following:

- Our learning algorithm focused on the dataset level. However, we found that there is still plenty of room for improvement if we dive one level deeper to the instance level. It would be interesting to develop and evaluate an algorithm that is capable of learning the best performing voting rule at the instance level.
- We focused on the aggregation sub-task of label ranking ensembles, while fixing the base label ranking algorithm to LRT or RPC. However, numerous label ranking algorithms were suggested in the literature, and to the best of our knowledge, an extensive comparison of these algorithms in the context of ensembles is lacking. Furthermore, it would be interesting to explore the option of combining label ranking models generated by different algorithms.
- Similarly, it would be interesting to further investigate the data sampling sub-task. All of the existing research on label ranking ensembles employed Bagging or Random Forest for data sampling. Future work may include developing and experimenting with other types of data sampling methods, such as Boosting. Moreover, the Stacking approach can be extended to support label ranking problems, by developing a meta label ranker that receives the output rankings of simple models as input attributes and returns a single aggregated ranking as output.
- We focused on the case of complete rankings. A compelling research direction would be to address the problem of incomplete rankings, as suggested by Aledo et al. (2017). While the approach and algorithms described in this paper can easily be extended to the case of incomplete rankings, an extensive evaluation is still required.

Declaration of competing interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Appendix. Additional experiments

We repeated the same set of analyses reported in Section 5 using the Ranking by Pairwise Comparison (RPC) algorithm, suggested by Hüllermeier et al. (2008), as the base label ranking learner, instead of the LRT algorithm which was used throughout Section 5. As we proceed to describe, our findings in the case of RPC were completely consistent with those of LRT.

Fig. 7 reports the average improvement (in percentages) of each of the voting rules in comparison to the single model approach. Similarly to the results reported in Section 5, the three variations of Plurality: Plurality, Veto and STV, obtained the poorest improvement results (in fact, they performed even worse than the single model approach) and Copeland obtained the best improvement result (better than that of Modal Ranking and Borda). Fig. 8 provides a breakdown of the 24 datasets, where each entry represents the rank of the given voting rule for the given dataset. As can be observed from the figure and in accordance with Section 5, there is no single best performing rule across all datasets.

Fig. 9 (top) shows the average improvement in percentages for each one of the seven ensembles based on voting rules as well as for the VRS algorithm and the Oracle (Dataset) method. Fig 9 (center) shows the percentage of datasets for which each aggregation method performed the best. Fig 9 (bottom) shows the average rank for each of the aggregation methods. Similarly to the results reported in Section 5, VRS outperforms all seven aggregation methods which are based on a single voting rule, including Modal Ranking, Borda and Copeland. Moreover, Oracle (Dataset) obtains a result which is only slightly higher than that obtained by Copeland, leaving a very slim margin for potential improvement.

Fig. 10 (top) shows the average improvement in percentages for each one of the seven ensembles based on voting rules as well as for VRS, Oracle (Dataset) and Oracle (Instance) methods. Fig. 10 (bottom) shows the percentage of instances for which each aggregation method performed the best. Consistent with the results reported in Section 5, we observe that choosing the right voting rule at the instance level, namely Oracle (Instance), can lead to additional improvement.

Fig. 11 shows the improvement in percentages as a function of the number of bags used for each of the aggregation methods. As can be seen in Fig. 11 and in accordance with Section 5, prediction performance grows with a diminishing effect with the number of simple models used.

In order to complete the picture, we performed the two sets of statistical significance tests described in Section 5.

In the first set of tests, we compared the seven aggregation methods based on single voting rules. The Friedman Aligned Ranks

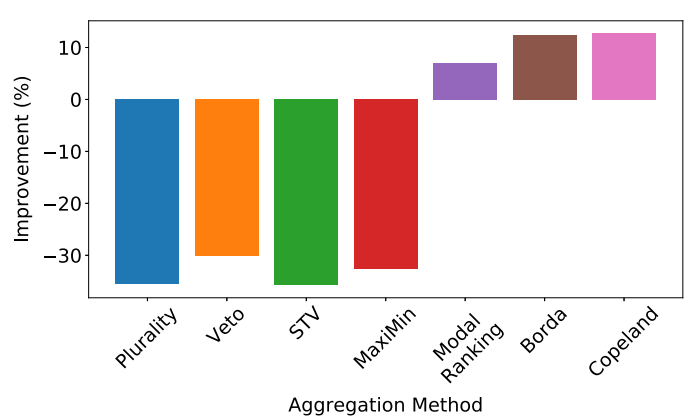

Fig. 7. RPC - Average improvement for ensembles based on the seven voting rules in comparison to the single model approach.

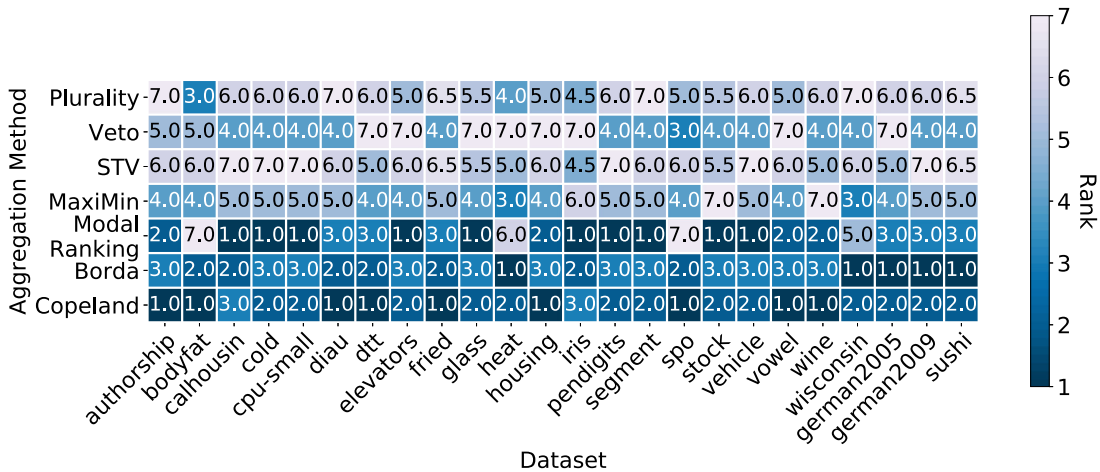

Fig. 8. RPC - Rank of voting rules ensembles for each of the 24 dataset.

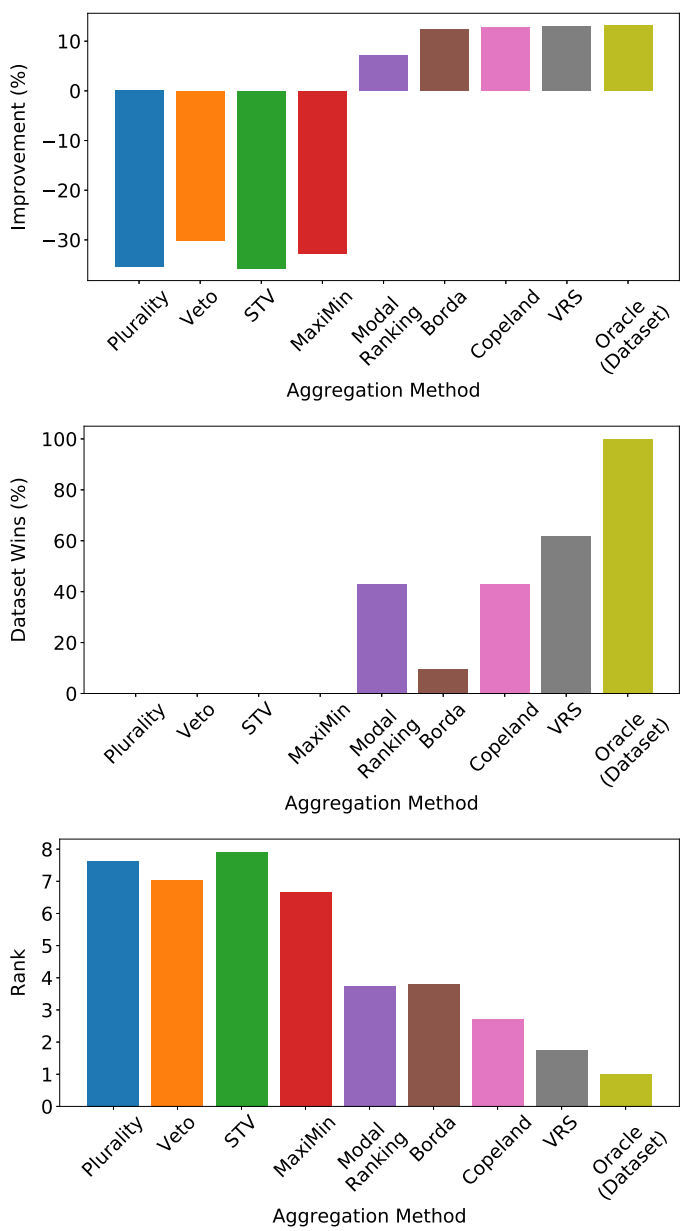

Fig. 9. RPC - Learning at the dataset level.

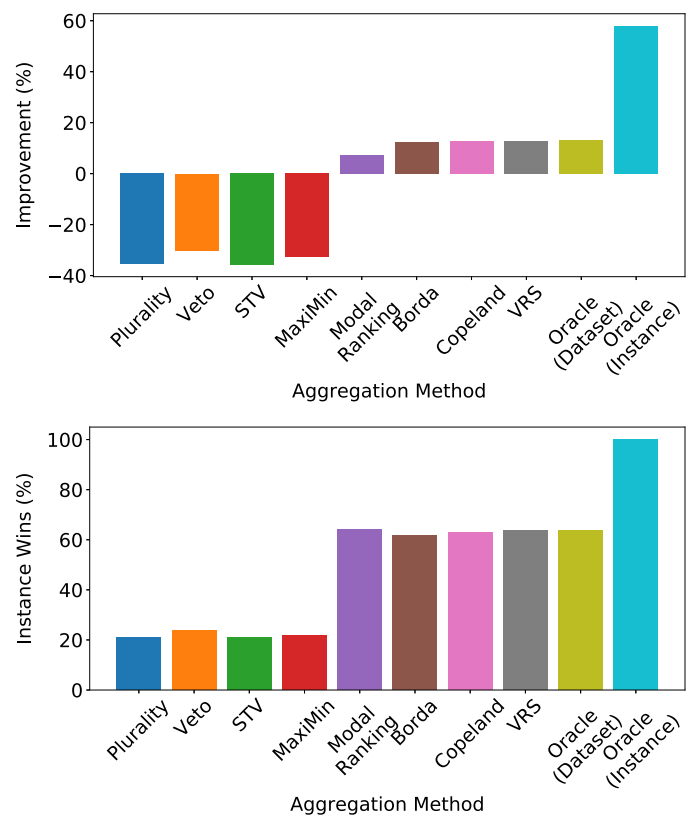

Fig. 10. RPC - The potential of learning at the instance level.

test with a confidence level of 95% rejected the null-hypothesis that all seven methods perform the same. Applying the Conover post hoc test supported our finding above: Copeland performed significantly better than the other six method, including Modal Ranking and Borda.

In the second set of tests, we compared the following four aggregation methods of interest: Copeland, VRS, Oracle (Dataset) and Oracle (Instance). The Friedman Aligned Ranks test with a confidence level of 95% rejected the null-hypothesis that all four methods perform the same. Applying the Conover post hoc test supported our findings above: VRS performed significantly better than Copeland; Oracle (Dataset) performed significantly better than VRS; and Oracle (Instance) performed significantly better than Oracle (Dataset).

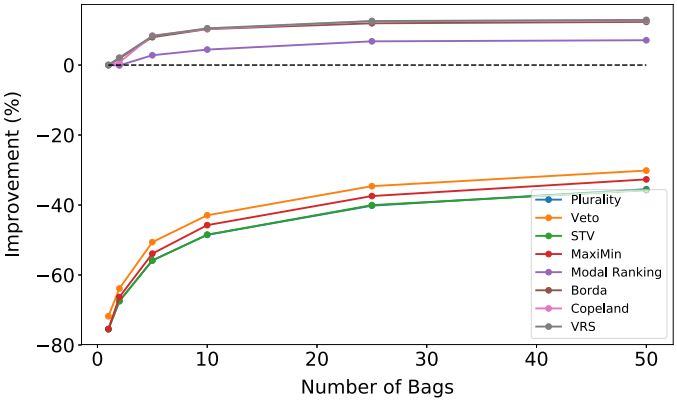

Fig. 11. RPC - Improvement in percentages as a function of the number of bags used.

Credit authorship contribution statement

Havi Werbin-Ofir: Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing - original draft. **Lihi Dery:** Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Erez Shmueli:** Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing.

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